

Assignment 1

You are provided with a data file named KIPRE_TNA_25-26_Final.csv, containing responses from a training needs assessment (TNA) survey conducted among KIPRE staff members recently.

The dataset consists of multiple sections and a large number of variables, some of which may be redundant, inconsistently structured, or require cleaning, recoding, merging, or exclusion prior to analysis.

You are expected to use any appropriate statistical software of their choice to explore the dataset and undertake necessary data preprocessing steps.

After data exploration and cleaning, the dataset should be used to address the questions provided below.

1. Which critical competency gaps exist across directorates, divisions, and job grades, and which areas require immediate training intervention?

Notes and hints about this question:

This question helps management identify - High-priority skill deficiencies - Departments with the greatest capacity gaps - Staff categories requiring urgent support	Likely outputs - Competency gap heat maps - Directorate ranking tables - Priority training matrix - Skill deficiency frequencies	Likely recommendations - Targeted capacity-building programmes - Directorate-specific training plans - Prioritized budget allocation
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2. To what extent are employees adequately prepared to meet emerging organizational, technological, and policy-related demands?

Hints

This question helps management identify - Future workforce readiness - Digital transformation preparedness - Emerging competency requirements - Institutional adaptability	Likely outputs - Emerging skills rankings - Workforce readiness summaries - Future skills dashboard - Trend analysis	Likely recommendations - Future-focused training programmes - Digital skills initiatives - Leadership and innovation training - Strategic workforce development planning
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The data has the following variables:

Variable name	Description	Section
SubmissionDate	Date the response was submitted	Metadata
start	Start time of form completion	Metadata
end	End time of form completion	Metadata
today	Date of response reference	Metadata
note_0	System note field	Metadata
note_1	System note field	Metadata
section_a.date_filling	Date of form filling	Section A
section_a.directorate	Respondent's directorate	Section A
section_a.division_section	Division or section of respondent	Section A
section_a.gender	Gender of respondent	Section A
section_a.age_bracket	Age category of respondent	Section A
section_a.officer_name	Name of officer	Section A
section_a.pf_no	Personnel file number	Section A

section_a.job_designation	Job title/designation	Section A
section_a.job_grade	Job grade level	Section A
section_a.length_service	Years of service	Section A
section_b.duties_responsibility_1–6	Reported duties and responsibilities (multiple items)	Section B
section_b.job_description	Job description text	Section B
section_b.role_clearly	Clarity of role definition	Section B
section_b.task_change	Frequency of task changes	Section B
section_c.rate_job_performance	Self-rated job performance	Section C
section_c.met_performance_targets	Achievement of performance targets	Section C
section_c.challenge	Main challenge faced	Section C
section_c.other_specify_challenge	Other challenges (open-ended)	Section C
section_d.tech_skill	Technical skills level	Section D
section_d.ict_skill	ICT skills level	Section D
section_d.comm_skill	Communication skills level	Section D
section_d.leadership_management	Leadership and management skills	Section D
section_d.policy_analysis_and_implementation	Policy analysis skills	Section D
section_d.finance_management	Financial management skills	Section D
section_d.customer_service	Customer service skills	Section D
section_d.monitoring_evaluation	Monitoring and evaluation skills	Section D
section_d.improve_skill	Skills improvement areas	Section D
section_e.attended_training	Whether training was attended	Section E
section_e.training_1_yes–6_yes	Training types attended (yes responses)	Section E
section_e.training_1_no–6_no	Training types not attended (no responses)	Section E
section_e.effective_training	Effectiveness of training	Section E
section_e.acquired_skill	Skills acquired from training	Section E
section_f.relevant_training_curr_role_1–6	Training relevance to current role	Section F
section_f.career_progression_training_1–6	Training relevance to career progression	Section F
section_f.level_training_needs	Level of training needs	Section F
section_f.training_type	Type of training required	Section F
section_g.training_mode1	Preferred training mode	Section G
section_g.preffered_duration	Preferred training duration	Section G
section_g.training_time	Preferred training time	Section G
section_h.support_training	Support needed for training	Section H
section_h.barrier	Barriers to training	Section H
section_h.others_support	Other support requirements	Section H
section_i.emerging_trends	Awareness of emerging trends	Section I
section_i.new_skill_you_need_1–6	New skills required	Section I
section_j.additional_comment	Additional comments	Section J
meta.instanceID	Unique response identifier	Metadata
KEY	Unique key identifier	Metadata
SubmitterID	ID of respondent	Metadata
SubmitterName	Name of respondent	Metadata
AttachmentsPresent	Whether attachments exist	Metadata
AttachmentsExpected	Expected attachments	Metadata

Status	Submission status	Metadata
ReviewState	Review status	Metadata
DeviceID	Device identifier used	Metadata
Edits	Number of edits made	Metadata
FormVersion	Version of data collection form	Metadata

Note: that this data should not be shared with anybody who is not part of this team. You can also use the sample of the report provided as your guide.